

Coconut oil enhances tomato carotenoid tissue accumulation compared to safflower oil in the Mongolian gerbil (*Meriones unguiculatus*).

Evidence suggests that monounsaturated and polyunsaturated fats facilitate greater absorption of carotenoids than saturated fats. However, the comparison of consuming a polyunsaturated fat source versus a saturated fat source on tomato carotenoid bioaccumulation has not been examined. The goal of this study was to determine the influence of coconut oil and safflower oil on tomato carotenoid tissue accumulation in Mongolian gerbils (*Meriones unguiculatus*) fed a 20% fat diet. Coconut oil feeding increased carotenoid concentrations among many compartments including total carotenoids in the serum ($p = 0.0003$), adrenal glandular phytoene ($p = 0.04$), hepatic phytofluene ($p = 0.0001$), testicular all-trans-lycopene ($p = 0.01$), and cis-lycopene ($p = 0.006$) in the prostate-seminal vesicle complex compared to safflower oil. Safflower oil-fed gerbils had greater splenic lycopene concentrations ($p = 0.006$) compared to coconut oil-fed gerbils. Coconut oil feeding increased serum cholesterol ($p = 0.0001$) and decreased hepatic cholesterol ($p = 0.0003$) compared to safflower oil. In summary, coconut oil enhanced tissue uptake of tomato carotenoids to a greater degree than safflower oil. These results may have been due to the large proportion of medium-chain fatty acids in coconut oil, which might have caused a shift in cholesterol flux to favor extrahepatic carotenoid tissue deposition.